MATHEMATICS II FOR DATA SCIENCE
GA-6 SOLUTIONS MAY 2024

---

1. Question

A function $T : V \to W$ between two vector spaces V and W is said to be a linear transformation if the following conditions hold:

Condition 1: $T(v_1 + v_2) = T(v_1) + T(v_2)$ for all $v_1, v_2 \in V$,
Condition 2: $T(cv_1) = cT(v_1)$ for all $v \in V$ and $c \in \mathbb{R}$

Consider the following function:

$$T : \mathbb{R}^2 \to \mathbb{R}$$
$$T(x, y) = \begin{cases} 3x & \text{if } y = 0 \\ 4y & \text{if } y \neq 0 \end{cases}$$

Which of the following statements is true?

- ☐ Condition 1 holds.
- ☐ Condition 1 does not hold.
- ☐ Condition 2 holds.
- ☐ Condition 2 does not hold.

1.1. Answers:

- Condition 1 does not hold.
- Condition 3 holds.

1.2. Solution

Let $v_1 = (x_1, 0)$, $v_2 = (x_2, 0)$, $v_3 = (x_3, y_3)$ and $x_4 = (x_4, y_4)$ where $x_1, x_2, x_3, x_4, y_3, y_4 \in \mathbb{R}$,$y_3 \neq 0$ and $y_4 \neq 0$.

Equality Check	LHS	RHS
$T(v_1 + v_2) = T(v_1) + T(v_2)$ $\therefore LHS = RHS$	$= T((x_1, 0) + (x_2, 0))$ $= T(x_1 + x_2, 0)$ $= 3x_1 + 3x_2$	$= T(x_1, 0) + T(x_2, 0)$ $= 3x_1 + 3x_2$

$T(v_3 + v_4) = T(v_3) + T(v_4)$ $\therefore LHS = RHS$	$= T((x_3, y_3) + (x_4, y_4))$ $= T(x_3 + x_4, y_3 + y_4)$ $= 4y_3 + 4y_4$	$= T(x_3, y_3) + T(x_4, y_4)$ $= 4y_3 + 4y_4$
$T(v_1 + v_3) = T(v_1) + T(v_3)$ $\therefore LHS \neq RHS$	$= T((x_1, 0) + (x_3, y3))$ $= T(x_1 + x_3, y_3)$ $= 4y_3$	$= T(x_1, 0) + T(x_3, y_3)$ $= 3x_1 + 4y_3$
$T(cv_1) = c \cdot T(v_1)$ $\therefore LHS = RHS$	$= T(c \cdot (x_1, 0)$ $= T(cx_1, 0)$ $= 3cx_1$	$= c \cdot T(x_1, 0)$ $= c \cdot (3x_1)$ $= 3cx_1$
$T(cv_3) = c \cdot T(v_3)$ $\therefore LHS = RHS$	$= T(c \cdot (x_3, y_3)$ $= T(cx_3, cy_3)$ $= 4cy_3$	$= c \cdot T(x_3, y_3)$ $= c \cdot (4y_3)$ $= 4cy_3$

From the above table, we can see that condition 2 hold under all possible circumstances. But, condition 1 fails to be true under certain cases. Hence, **option 2** and **option 3** are correct, and **option 1** and **option 4** are incorrect.

---

2. Question

Suppose the matrix representation of a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ with respect to ordered bases $\beta = \{(1,0,1), (0,1,0), (0,0,1)\}$ for the domain and $\gamma = \{(1,0,0), (0,1,0), (1,0,1)\}$ for the range, is $I_{3\times3}$,i.e., the identity matrix of order 3. Let A denote the matrix representation of the linear transformation T with respect to the standard ordered basis of $\mathbb{R}^3$ for both domain and range. Which of the following are true?

- ☐ $A = I_{3\times3}$ i.e., identity matrix of order 3.
- ☐ A is a singular matrix
- ☐ $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$
- ☐ $det(A) = 1.$
- ☐ $det(A) = -1.$
- ☐ $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

2.1. Answers :

- $det(A) = 1$
- $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

2.2. Solution

It is given that $I_{3\times3}$ is the matrix representation of $T : \mathbb{R}^3 \to \mathbb{R}^3$ with respect to β and γ. With this information, we can find the following:

$$\begin{aligned} T(1,0,1) &= 1 \cdot (1,0,0) + 0 \cdot (0,1,0) + 0 \cdot (1,0,1) \\ T(1,0,1) &= (1,0,0) \end{aligned} \tag{1}$$

$$\begin{aligned} T(0,1,0) &= 0 \cdot (1,0,0) + 1 \cdot (0,1,0) + 0 \cdot (1,0,1) \\ T(0,1,0) &= (0,1,0) \end{aligned} \tag{2}$$

$$\begin{aligned} T(0,0,1) &= 0 \cdot (1,0,0) + 0 \cdot (0,1,0) + 1 \cdot (1,0,1) \\ T(0,0,1) &= (1,0,1) \end{aligned} \tag{3}$$

Any vector $(x, y, z) \in \mathbb{R}^3$ can be represented as a linear combination of the basis vectors in β.

$$\begin{aligned} (x,y,x) &= p \cdot (1,0,1) + q \cdot (0,1,0) + r \cdot (0,0,1) \\ (x,y,z) &= (p, q, p+r) \end{aligned}$$

Here, $p, q, r \in \mathbb{R}$. From the above equation, we can see that $p = x$, $q = y$ and $r = z - x$.

$$(x,y,z) = x \cdot (1,0,1) + y \cdot (0,1,0) + (z-x) \cdot (0,0,1)$$

We can write $T(x, y, z)$ as : –

$$\begin{aligned} T(x,y,z) &= T(x \cdot (1,0,1) + y \cdot (0,1,0) + (z-x) \cdot (0,0,1)) \\ T(x,y,z) &= x \cdot T(1,0,1) + y \cdot T(0,1,0) + (z-x) \cdot T(0,0,1) \end{aligned}$$

From (1), (2) and (3), we get : –

$$\begin{aligned} T(x,y,z) &= x \cdot (1,0,0) + y \cdot (0,1,0) + (z-x) \cdot T(1,0,1) \\ T(x,y,z) &= (z, y, z-x) \end{aligned}$$

Now, we have to find matrix representation with respect to the standard ordered basis.

$$\begin{aligned} T(1,0,0) &= (0,0,-1) \\ T(1,0,0) &= 0 \cdot (1,0,0) + 0 \cdot (0,1,0) - 1(0,0,1) \end{aligned}$$

$$\begin{aligned} T(0,1,0) &= (0,1,0) \\ T(0,1,0) &= 0 \cdot (1,0,0) + 1 \cdot (0,1,0) + 0(0,0,1) \end{aligned}$$

$$T(0,0,1) = (1,0,1)$$
$$T(0,0,1) = 1 \cdot (1,0,0) + 0 \cdot (0,1,0) + 1(0,0,1)$$

The matrix representation of $T : \mathbb{R}^3 \to \mathbb{R}^3$ with respect to the standard ordered basis is : –

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Hence, **option 6** is correct. **Option 1** and **option 3** are incorrect.

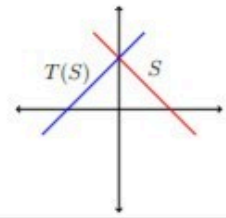
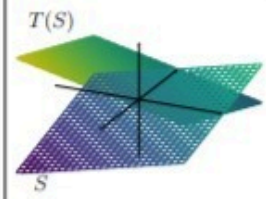
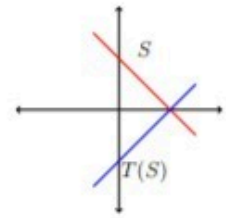
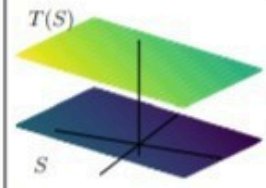
$$det(A) = 0 \cdot (1-0) - 0 \cdot (0-0) + 1 \cdot (0-(-1))$$
$$det(A) = 1$$

Hence, **option 4** is correct. **Option 2** and **option 5** are incorrect.

---

3. Question

Match the linear transformations and sets of vectors in column A with the images of those sets under the linear transformation in column B and the geometric representations of both sets in column C.

	Matrix form of linear transformation (Column A)		Image of the given set (Column B)		Geometric representations (Column C)
i)	$T : \mathbb{R}^3 \to \mathbb{R}^3$ $T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 2 & 3 \end{bmatrix}$ Set: $S = \{(x, y, z) \mid x + y + z = 1\}$	a)	$T(S) = \{(x, y) \mid x - y = 1\}$	1)	
ii)	$T : \mathbb{R}^3 \to \mathbb{R}^3$ $T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ Set: $S = \{(x, y, z) \mid x + y + z = 1\}$	b)	$T(S) = \{(x, y, z) \mid x + y + z = 3\}$	2)	
iii)	$T : \mathbb{R}^2 \to \mathbb{R}^2$ $T = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ Set: $S = \{(x, y) \mid x + y = 1\}$	c)	$T(S) = \{(x, y) \mid x - y = -1\}$	3)	
iv)	$T : \mathbb{R}^2 \to \mathbb{R}^2$ $T = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ Set: $S = \{(x, y) \mid x + y = 1\}$	d)	$T(S) = \{(x, y, z) \mid x + y - z = 1\}$	4)	

Choose the correct option from the following.

- [ ] i $\rightarrow$ d $\rightarrow$ 2, ii $\rightarrow$ b $\rightarrow$ 4.
- [ ] i $\rightarrow$ b $\rightarrow$ 4, ii $\rightarrow$ d $\rightarrow$ 2.
- [ ] iii $\rightarrow$ a $\rightarrow$ 1, iv $\rightarrow$ c $\rightarrow$ 3
- [ ] iii $\rightarrow$ a $\rightarrow$ 3, iv $\rightarrow$ c $\rightarrow$ 1

3.1. Answers :

- i $\rightarrow$ b $\rightarrow$ 4, ii $\rightarrow$ d $\rightarrow$ 2
- iii $\rightarrow$ a $\rightarrow$ 3, iv $\rightarrow$ c $\rightarrow$ 1

3.2. Solution

<u>Column A</u> (i)

$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$. Matrix representation $T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 2 & 3 \end{bmatrix}$.

Using standard ordered basis for both domain and codomain, we can recover $T(x, y, z)$.

$$\begin{aligned} T(1,0,0) &= 1 \cdot (1,0,0) + 0 \cdot (0,1,0) + 2 \cdot (0,0,1) \\ T(1,0,0) &= (1,0,2) \end{aligned} \tag{1}$$

$$\begin{aligned} T(0,1,0) &= 0 \cdot (1,0,0) + 1 \cdot (0,1,0) + 2 \cdot (0,0,1) \\ T(0,1,0) &= (0,1,2) \end{aligned} \tag{2}$$

$$\begin{aligned} T(0,0,1) &= 0 \cdot (1,0,0) + 0 \cdot (0,1,0) + 3 \cdot (0,0,1) \\ T(0,0,1) &= (0,0,3) \end{aligned} \tag{3}$$

From (1), (2) and (3) we can recover $T(x, y, z)$

$$\begin{aligned} (x,y,z) &= x \cdot (1,0,0) + y \cdot (0,1,0) + z \cdot (0,0,1) \\ T(x,y,z) &= T(x \cdot (1,0,0) + y \cdot (0,1,0) + z \cdot (0,0,1)) \\ T(x,y,z) &= x \cdot T(1,0,0) + y \cdot T(0,1,0) + z \cdot T(0,0,1) \\ T(x,y,z) &= x \cdot (1,0,2) + y \cdot (0,1,2) + z \cdot (0,0,3) \\ T(x,y,z) &= (x, y, 2x + 2y + 3z) \end{aligned}$$

$S = \{(x, y, z) | x + y + z = 1\}$. This set can be written as $S = \{(x, y, 1 - x - y)\}$.
Every vector in set S is of the form $(x, y, 1 - x - y)$.

$$T(x, y, 1 - x - y) = (x, y, 2x + 2y + 3 \cdot (1 - x - y))$$

$$T(x, y, 1-x-y) = (x, y, 2x+2y+3-3x-3y)$$
$$T(x, y, 1-x-y) = (x, y, 3-x-y)$$

Every vector in set $T(S)$ is of the form $(x, y, 3-x-y)$.
$T(S) = \{(x, y, 3-x-y)$
$T(S) = \{(x, y, z)|z = 3-x-y\}$
$T(S) = \{(x, y, z)|x+y+z = 3\}$

Recall, $S = \{(x, y, z)|x+y+z = 1\}$

Every vector in set S will lie on the plane $x+y+z = 1$ and every vector in set $T(S)$ will lie on the plane $x+y+z = 3$. A point of the form (x, y, z) cannot satisfy both $x+y+z = 1$ and $x+y+z = 3$. Therefore, both the planes are parallel to each other.

Therefore, $i \to b \to 4$ is correct.

<u>Column A</u> (ii)

$T : \mathbb{R}^3 \to \mathbb{R}^3$. Matrix representation $T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$.

Using standard ordered basis for both domain and codomain, we can recover $T(x, y, z)$.

$$T(1, 0, 0) = 1 \cdot (1, 0, 0) + 0 \cdot (0, 1, 0) + 0 \cdot (0, 0, 1)$$
$$T(1, 0, 0) = (1, 0, 0) \qquad (4)$$

$$T(0, 1, 0) = 0 \cdot (1, 0, 0) + 1 \cdot (0, 1, 0) + 0 \cdot (0, 0, 1) \qquad (5)$$
$$T(0, 1, 0) = (0, 1, 0)$$

$$T(0, 0, 1) = 0 \cdot (1, 0, 0) + 0 \cdot (0, 1, 0) - 1 \cdot (0, 0, 1) \qquad (6)$$
$$T(0, 0, 1) = (0, 0, -1)$$

From (4), (5) and (6) we can recover $T(x, y, z)$

$$(x, y, z) = x \cdot (1, 0, 0) + y \cdot (0, 1, 0) + z \cdot (0, 0, 1)$$
$$T(x, y, z) = T(x \cdot (1, 0, 0) + y \cdot (0, 1, 0) + z \cdot (0, 0, 1))$$
$$T(x, y, z) = x \cdot T(1, 0, 0) + y \cdot T(0, 1, 0) + z \cdot T(0, 0, 1)$$
$$T(x, y, z) = x \cdot (1, 0, 0) + y \cdot (0, 1, 0) + z \cdot (0, 0, -1)$$
$$T(x, y, z) = (x, y, -z)$$

$S = \{(x, y, z)|x+y+z = 1\}$. This set can be written as $S = \{(x, y, 1-x-y)\}$.
Every vector in set S is of the form $(x, y, 1-x-y)$.

$$T(x, y, 1-x-y) = (x, y, -(1-x-y))$$
$$T(x, y, 1-x-y) = (x, y, x+y-1)$$

Every vector in set $T(S)$ is of the form $(x, y, x+y-1)$
$T(S) = \{(x, y, x+y-1)\}$
$T(S) = \{(x, y, z) | z = x+y-1\}$
$T(S) = \{(x, y, z) | x+y-z = 1\}$

Every vector in set S will lie on the plane $x+y+z = 1$ and every vector in set $T(S)$ will lie on the plane $x+y-z = 1$. These two planes will intersect to form a straight line in $\mathbb{R}^3$.

Therefore $ii \to d \to 2$ is correct.
Hence, **option 1** is incorrect and **option 2** is correct.

<u>Column A</u> (iii)

$T : \mathbb{R}^3 \to \mathbb{R}^3$. Matrix representation $T = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.
Using standard ordered basis for both domain and codomain, we can recover $T(x, y)$.

$$\begin{aligned} T(1,0) &= 1 \cdot (1,0) + 0 \cdot (0,1) \\ T(1,0) &= (1,0) \end{aligned} \tag{7}$$

$$\begin{aligned} T(0,1) &= 0 \cdot (1,0) - 1 \cdot (0,1) \\ T(0,1) &= (0,-1) \end{aligned} \tag{8}$$

From (7) and (8) we can recover $T(x, y)$

$$\begin{aligned} (x,y) &= x \cdot (1,0) + y \cdot (0,1) \\ T(x,y) &= T(x \cdot (1,0) + y \cdot (0,1)) \\ T(x,y) &= x \cdot T(1,0) + y \cdot T(0,1) \\ T(x,y) &= x \cdot (1,0) + y \cdot (0,-1) \\ T(x,y) &= (x,-y) \end{aligned}$$

$S = \{(x, y) | x+y = 1\}$. This can be written as $S = \{(x, 1-x)\}$. Every vector in set S is of the form $(x, 1-x)$.

$$\begin{aligned} T(x,1-x) &= (x,-(1-x)) \\ T(x,1-x) &= (x,x-1) \end{aligned}$$

Every vector in $T(S)$ is of the form $(x, 1-x)$.
$T(S) = \{(x, x-1)\}$
$T(S) = \{(x, y) | y = x-1\}$
$T(S) = \{(x, y) | x-y = 1\}$

Every vector in set S will lie on the line $x+y = 1$ and every vector in set $T(S)$ will lie on the line $x-y = 1$. Both the lines intersect at $(1, 0)$.

Therefore, $iii \to a \to 3$ is correct.

<u>Column A</u> (iv)

$T : \mathbb{R}^3 \to \mathbb{R}^3$. Matrix representation $T = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$.

Using standard ordered basis for both domain and codomain, we can recover $T(x, y)$.

$$\begin{aligned} T(1,0) &= 1 \cdot (1,0) + 2 \cdot (0,1) \\ T(1,0) &= (1,2) \end{aligned} \tag{9}$$

$$\begin{aligned} T(0,1) &= 0 \cdot (1,0) + 1 \cdot (0,1) \\ T(0,1) &= (0,1) \end{aligned} \tag{10}$$

From (7) and (8) we can recover $T(x, y)$

$$\begin{aligned} (x,y) &= x \cdot (1,0) + y \cdot (0,1) \\ T(x,y) &= T(x \cdot (1,0) + y \cdot (0,1)) \\ T(x,y) &= x \cdot T(1,0) + y \cdot T(0,1) \\ T(x,y) &= x \cdot (1,2) + y \cdot (0,1) \\ T(x,y) &= (x, 2x + y) \end{aligned}$$

$S = \{(x, y) | x + y = 1\}$. This can be written as $S = \{(x, 1 - x)\}$. Every vector in set S is of the form $(x, 1 - x)$.

$$\begin{aligned} T(x, 1-x) &= (x, 2x + 1 - x) \\ T(x, 1-x) &= (x, 1 + x) \end{aligned}$$

Every vector in $T(S)$ is of the form $(x, 1 + x)$.
$T(S) = \{(x, 1 + x)\}$
$T(S) = \{(x, y) | y = 1 + x\}$
$T(S) = \{(x, y) | x - y = -1\}$

Every vector in set S will lie on the line $x + y = 1$ and every vector in set $T(S)$ will lie on the line $x - y = -1$. Both the lines intersect at $(0, 1)$.

Therefore, $iv \to c \to 1$ is correct.
Hence, **option 3** is incorrect and **option 4** is correct.

---

4. Question

Consider a linear transformation $S : M_2(\mathbb{R}) \to M_2(\mathbb{R})$ such that $S(A) = A^T$. Let B be the matrix representation of S with respect to the ordered bases:

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

of $M_2(\mathbb{R})$. Choose the set of correct options:

- ☐ The order of the matrix B is 2×2.
- ☐ The order of the matrix B is 4×4.
- ☐ The dimension of the row space of the the matrix B is 4.
- ☐ The dimension of the column space of the matrix B is 3.
- ☐ The nullity of the the matrix B is 1.
- ☐ The rank of the the matrix B is 4.
- ☐ S is surjective.

4.1. Answers :

- The order of the matrix B is 4×4.
- The dimension of the row space if B is 4.
- The rank of B is 4.
- S is surjective.

4.2. Solution

The order of the matrix representation of a linear transformation will be $m \times n$, where m is the cardinality of the basis of co-domain and n is the cardinality of the basis of domain.
The cardinality of the basis of both domain and codomain is 4. Therefore, the order of the matrix representation of S will be 4×4. Hence, **option 2** is correct and **option 1** is incorrect.

It is given that $S(A) = A^T$.

$$S\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$
$$S\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) = 1 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$S\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$
$$S\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) = 0 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 1 \cdot \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$S\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$
$$S\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) = 0 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 1 \cdot \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$S\left(\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$S\left(\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right) = 0 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 1 \cdot \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

The matrix representation of S with respect to the ordered basis is $B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

The reduced row echelon form of B is $I_{4\times4}$, where I is the identity matrix.
Since there will be no column in $I_{4\times4}$ without a leading entry, the rank of B is 4 and the nullity of B is 0.
Since the row rank of a matrix is equal to its column rank, we can say that the dimension of the row space is 4. Hence, **option 3** and **option 6** are correct. **Option 4** and **option 5** are incorrect.

Since all the columns of $I_{4\times4}$ have leading entries, we can say that the dimension of $Img(S)$ will be 4.

Notice that the cardinality of the basis set of co-domain is also four. Thus we can say $dim(Img(S)) = dim(Co-domain)$. Therefore, S is surjective. Hence, **option 7** is correct.

---

5. Question

Consider the following statements:

Statement 1: Consider a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^4$ such that T is not injective. Then $rank(T) < 3$.

Statement 2: If $T : V \to W$ is a linear transformation, whose matrix representation with respect to some ordered bases is given by the matrix $\begin{bmatrix} 0 & \alpha & \gamma \\ 1 & 0 & \gamma \\ 0 & \beta & \frac{\gamma\beta}{\alpha} \end{bmatrix}$, where $\alpha, \beta, \gamma \in \mathbb{R} \setminus \{0\}$, then the rank of the linear transformation T is 3.

Statement 3: If $T : \mathbb{R}^3 \to \mathbb{R}^4$ is a linear transformation such that $T(x, y, z) = (2x - z, 3y - 2z, z, 0)$, then $\{(-3, 1, 1, 0), (1, -5, 1, 0), (3, 5, -1, 0)\}$ is a basis of the image space.

Statement 4: If $T : M_2(\mathbb{R}) \to M_2(\mathbb{R})$ is a linear transformation such that $T(A) = PA$, where $A \in M_2(\mathbb{R})$ and $P = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$, then $\left\{\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right\}$ is a basis of the kernel.

Write down the statement numbers corresponding to the correct statements in increasing order.

[Note: Suppose Statement 1, Statement 2, and Statement 4 are correct then your answer should be 124. Similarly, if Statement 2 and Statement 3 are correct then your answer should be 23. In this list one or more than one statement can be correct. Do not add any space between the digits.]

5.1. Answer : 13

5.2. Solution

<u>Statement - 1</u>

From the Rank-Nullity Theorem, we can say that $rank(T) + nullity(T) = dim\left(\mathbb{R}^3\right)$, where $\mathbb{R}^3$ is the domain of T.

Since T is not injective, $nullity(T) \geqslant 1$. Let us assume that $nullity(T) = 1$. Then $rank(T) = 3 - 1 = 2$, where 3 is the dimension of $\mathbb{R}^3$.
From this we can see that $rank(T) \leqslant 2$. Hence, **statement 1** is correct.

<u>Statement - 2</u>

The matrix representation of T is $M = \begin{bmatrix} 0 & \alpha & \gamma \\ 1 & 0 & \gamma \\ 0 & \beta & \dfrac{\gamma\beta}{\alpha} \end{bmatrix}$. The reduced row echelon form of this matrix is

$R = \begin{bmatrix} 1 & 0 & \gamma \\ 0 & 1 & (\gamma/\alpha) \\ 0 & 0 & 0 \end{bmatrix}$. There are only two columns with leading entries in R. Therefore, $rank(T) = 2$. Hence, **statement 2** is false.

<u>Statement - 3</u>

Every vector in $Img(T)$ will be of the form $(2x - z, 3y - 2z, z, 0)$.
For the set $\{(-3, 1, 1, 0), (1, -5, 1, 0), (3, 5, -1, 0)\}$ to be a basis for $Img(T)$, the must be a unique solution for $Ax = q$, where

$$A = \begin{bmatrix} -3 & 1 & 3 \\ 1 & -5 & 5 \\ 1 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, q = \begin{bmatrix} 2x - z \\ 3y - 2z \\ z \\ 0 \end{bmatrix}$$

Unique solution would also imply that the columns of A are linearly independent. Linear independence and unique solution $Ax = q$ is necessary and sufficient for the given set to be a basis set for $Img(T) = \{(2x - z, 3y - 2z, z, 0) | x, y, z \in R\}$.

Let us augment A and q,

$$(A|q) = \left[\begin{array}{ccc|c} -3 & 1 & 3 & 2x - z \\ 1 & -5 & 5 & 3y - 2z \\ 1 & 1 & -1 & z \\ 0 & 0 & 0 & 0 \end{array}\right]$$

For the ease of calculation, let us take $a = 2x - z, b = 3y - 2z$ and $c = z$.
Let us apply elementary row operations on $(A|q)$,

$$\left[\begin{array}{ccc|c} -3 & 1 & 3 & a \\ 1 & -5 & 5 & b \\ 1 & 1 & -1 & c \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R3-R2} \left[\begin{array}{ccc|c} -3 & 1 & 3 & a \\ 1 & -5 & 5 & b \\ 0 & 6 & -6 & c-b \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{\frac{R3}{6}} \left[\begin{array}{ccc|c} -3 & 1 & 3 & a \\ 1 & -5 & 5 & b \\ 0 & 1 & -1 & (c-b)/6 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{R2+5R3} \left[\begin{array}{ccc|c} -3 & 1 & 3 & a \\ 1 & 0 & 0 & (b-5c)/6 \\ 0 & 1 & -1 & (c-b)/6 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R1-R3} \left[\begin{array}{ccc|c} -3 & 0 & 4 & (6a-b+5c)/6 \\ 1 & 0 & 0 & (b-5c)/6 \\ 0 & 1 & -1 & (c-b)/6 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{R1+3R2}\left[\begin{array}{ccc|c} 0 & 0 & 4 & (6a-4b+8c)/6 \\ 1 & 0 & 0 & (b-5c)/6 \\ 0 & 1 & -1 & (c-b)/6 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{\frac{R1}{4}}\left[\begin{array}{ccc|c} 0 & 0 & 1 & (6a-4b+8c)/24 \\ 1 & 0 & 0 & (b-5c)/6 \\ 0 & 1 & -1 & (c-b)/6 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{R3+R1}\left[\begin{array}{ccc|c} 0 & 0 & 1 & (6a-4b+8c)/24 \\ 1 & 0 & 0 & (b-5c)/6 \\ 0 & 1 & 0 & (6a-8b+12c)/6 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R1\leftrightarrow R2}\left[\begin{array}{ccc|c} 1 & 0 & 0 & (b-5c)/6 \\ 0 & 0 & 1 & (6a-4b+8c)/24 \\ 0 & 1 & 0 & (6a-8b+12c)/6 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{R2\leftrightarrow R3}\left[\begin{array}{ccc|c} 1 & 0 & 0 & (b-5c)/6 \\ 0 & 1 & 0 & (6a-8b+12c)/6 \\ 0 & 0 & 1 & (6a-4b+8c)/24 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The above matrix is in reduced row echelon form. We can see that all the three columns have leading entries. So all the unknown variables are dependent variables. We can also see that there is no row of the form $[0, 0, 0|p]$ where $p \in \mathbb{R}$ and $p \neq 0$. So the system of linear equations is consistent. A system of linear equations will have a unique solution if it is consistent and has no independent variables. Therefore, we have a unique solution for $Ax = b$. We can substitute the original values of a, b and c to see that Ax is indeed equal to b.

Therefore, the set $\{(-3, 1, 1, 0), (1, -5, 1, 0), (3, 5, -1, 0)\}$ can be a basis set for $Img(T)$. Hence, **statement 3** is correct.

<u>Statement - 4</u>

It is given that $T(A) = PA,$ where $A \in M_{2\times 2}(\mathbb{R})$ and $P = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$.

$Ker(T) = \left\{B|T(B) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}\right\}$, where $Ker(T)$ refers to the kernel of T. In simple words, multiplying matrix P with any matrix in $Ker(T)$ will result in $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

Let us assume that all the matrices in $Ker(T)$ is of the form B, where $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where $a, b, c, d \in \mathbb{R}$ and $PB = 0$.

$$PB = 0$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \times \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} c-a & d-b \\ a-c & b-d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (1)$$

From (1), we can see that $a = c$ and $b = d$.

B is of the form $\begin{bmatrix} a & b \\ a & b \end{bmatrix}$. Every matrix in $Ker(T)$ should be of the form $\begin{bmatrix} a & b \\ a & b \end{bmatrix}$, where $a, b \in \mathbb{R}$.

Statement-4 claims $\left\{\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right\}$ to be the basis for $Ker(T)$. However $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ is not of the form $\begin{bmatrix} a & b \\ a & b \end{bmatrix}$, where $a, b \in \mathbb{R}$. Hence, **statement-4** is incorrect.

Comprehension $(6, 7, 8)$

Suppose a bread-making machine B makes 6 breads from 2 eggs, 3 (in hundreds) grams of wheat, and 1 (in hundred) grams of sugar. B also makes 8 breads from 3 eggs, 4 (in hundreds) grams of wheat, and 2 (in hundreds) grams of sugar, and 10 breads from 5 eggs, 5 (in hundreds) grams of wheat, and 3 (in hundreds) grams of sugar. Suppose the production of breads is a linear function of the amount of eggs, wheat (in hundreds), and sugar (in hundreds) used as raw ingredients. Based on the above data answer the following questions. Suppose x eggs, y (in hundreds) grams of wheat, and z (in hundreds) grams of sugar are used as the raw materials to produce $ax + by + cz$ number of breads. We can express this as follows:

$$T : \mathbb{R}^3 \to \mathbb{R}$$
$$T(x, y, z) = ax + by + cz$$

where the co-ordinates in $\mathbb{R}^3$ denote the number of eggs, amount (in grams) of wheat (in hundreds), and amount (in grams) of sugar (in hundreds). Observe that T is a linear transformation.

Use the above information for Question 6,7 and 8.

---

6. Question

Choose the correct set of options from the following. (MSQ)

- ☐ $Nullity(T) = 1$
- ☐ $Rank(T) = 1$
- ☐ $Nullity(T) = 2$
- ☐ $Rank(T) = 2$
- ☐ $Nullity(T) = 3$
- ☐ $Rank(T) = 3$
- ☐ T is neither one to one nor onto.
- ☐ T is one to one but not onto.
- ☐ T is onto but not one to one.
- ☐ T is an isomorphism.

6.1. Answers :

- $Rank(T) = 1$
- $Nullity(T) = 2$
- T is onto but not one to one.

6.2. Solution

$T(x, y, z) = ax + by + cz$. From the given information we can get the following linear equations.

$$\begin{aligned} T(2,3,1) &= 6 \\ T(2,3,1) &= 2a + 3b + c \\ 2a + 3b + c &= 6 \qquad (1) \end{aligned}$$

$$\begin{aligned} T(3,4,2) &= 8 \\ T(3,4,2) &= 3a + 4b + 2c \\ 3a + 4b + 2c &= 8 \qquad (2) \end{aligned}$$

$$\begin{aligned} T(5,5,3) &= 10 \\ T(5,5,3) &= 5a + 5b + 3c \\ 5a + 5b + 3c &= 10 \qquad (3) \end{aligned}$$

Let us covert the linear equation (1), (2) and (3) into their matrix form $Ax = b$, where

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 4 & 2 \\ 5 & 5 & 3 \end{bmatrix}, x = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, b = \begin{bmatrix} 6 \\ 8 \\ 10 \end{bmatrix}$$

Let us augment A and b.

$$(A|b) = \left[\begin{array}{ccc|c} 2 & 3 & 1 & 6 \\ 3 & 4 & 2 & 8 \\ 5 & 5 & 3 & 10 \end{array}\right]$$

The reduced row echelon form of $(A|b)$ is $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 \end{array}\right]$. From this, we get $a = 0, b = 2$ and $c = 0$.

Therefore, $T(x, y, z) = 2y$.
We can see that the number of breads depends only on the amount (in grams) of wheat (in hundreds).

$T(x, y, z)$ will be equal to 0 whenever $y = 0$. Therefore, every vector in the kernel of T will be of the form $(x, 0, z) = x \cdot (1, 0, 0) + z \cdot (0, 0, 1)$. Notice that $\{(1, 0, 0), (0, 0, 1)\}$ is a linearly independent set. Therefore, the basis for the kernel of T is $\{(1, 0, 0), (0, 0, 1)\}$.

Thus we can conclude that $Nullity(T) = 2$. A linear transformation will be one-to-one if and only if its nullity is equal to 0. Hence, **option 3** is correct. **Option 1**, **option 5** and **option 8** are incorrect.

Rank-Nullity Theorem for a linear transformation S: $Rank(S) + Nullity(S) = dim(Domain)$.
From the Rank-Nullity Theorem, we can see that $Rank(T) = 1$. A linear transformation is onto if and only if its rank is equal to the dimension of the co-domain. Here, the codomain is $\mathbb{R}$ and $dim(\mathbb{R}) = 1 = Rank(T)$. Therefore, T is onto.

Since T is onto but not one-to-one. we can conclude that T is not an isomorphism.

Hence, **option 2** and **option 9** are correct. **Option 4**, **option 6**, **option 7** and **option 10** are incorrect.

---

7. Question

Choose the set of correct statements. (MSQ)

- ☐ If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then 9 breads are produced.
- ☐ If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then no bread is produced.
- ☐ If only 3 (in hundreds) grams of wheat is used, then 6 breads are produced.
- ☐ If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then no bread is produced.
- ☐ If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then 6 breads are produced.

7.1. Answers :

- If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then no bread is produced.
- If only 3 (in hundreds) grams of wheat is used, then 6 breads are produced.
- If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then 6 breads are produced.

7.2. Solution

We know that $T(x, y, z) = 2y$. If wheat is not used, then no bread will produced. Hence **option 1** is incorrect and **option 2** is correct.

$T(0, 3, 0) = 6$. Hence, **option 3** is correct.

$T(0, 3, 1) = 6$. Hence, **option 4** is incorrect and **option 5** is correct.

---

8. Question

How many breads are produced by the machine from 6 eggs, 10 (in hundreds) grams of wheat, and 5 (in hundreds) grams of sugar? (NAT)

8.1. Answer : 20

8.2. Solution

$T(6, 10, 5) = 2(10) = 20$. Therefore, 20 breads will be produced by using 6 eggs,10 (in hundreds) grams of wheat, and 5 (in hundreds) grams of sugar.

---

9. Question

Let T be a linear transformation from $\mathbb{R}^8$ to $\mathbb{R}^7$. Suppose a basis for the null space of T has 6 vectors. How many linearly independent vectors are needed to form a basis for the range of T?

9.1. Answer : 2

9.2. Solution

$dim(range\ of\ T) = Rank(T)$.
If the basis of the null space of T has 6 vectors, then $Nullity(T) = 6$.
Rank-Nullity Theorem for a linear transformation S: $Rank(S) + Nullity(S) = dim(Domain)$.
From the Rank-Nullity Theorem, we can see that:

$$\begin{aligned} Rank(T) + 6 &= dim\left(\mathbb{R}^8\right) \\ Rank(T) &= 8 - 6 \\ Rank(T) &= 2 \end{aligned}$$

Therefore, we need 2 linearly independent vectors to form a basis for the range of T.

---

10. Question

Let M be the set of all skew-symmetric matrices of order 10. Then M forms a vector space under matrix addition and scalar multiplication. Let T be a linear transformation from M to R defined by $T(A) = 8\ tr(A)$, where $tr(A) = trace\ of\ A$. What is the nullity of T?

10.1. Answer : 45

10.2. Solution

M is the set of all skew-symmetric matrices of order 10.
We can write $M = \left\{A | A \in M_{10\times10}(\mathbb{R}), A = -A^T\right\}$. Notice that while performing matrix transpose, the diagonal elements will not change their positions. However, the diagonal elements will have to be equal to its negative. This is possible only if all the diagonal elements are equal to 0.

The sum of the diagonal elements of a matrix is its trace. Trace of any Skew-Symmetric matrix will be 0.

$T(A) = 0$ for all $A \in M$. Therefore, $Nullity(T) = dim(M)$. The basis set of M will have 44 linearly independent matrices.

Let us take a 3×3 Skew-Symmetric matrix. $A = \begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ b & -c & 0 \end{bmatrix}$, where $a, b, c \in \mathbb{R}$.
Notice that $A = -A^T$. Every 3×3 skew-symmetric matrix will be a scalar multiple of A.
A can be written as $a \cdot \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + b \cdot \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} + c \cdot \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$.
$\left\{\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}\right\}$ is a linearly independent set. It can be a basis set for the set of all 3×3 skew-symmetric matrices. Therefore, the dimension of the set of all 3×3 skew-symmetric matrices is 3.

The number of unique variables in $\begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ b & -c & 0 \end{bmatrix}$ is also 3. They are a, b and c. All the three are to the right of the diagonal entries.

A 10×10 skew-symmetric matrix will have 44 such unique variables.
In the first row, there will be 9 such unique variables, as 9 entries are to the right of the diagonal.
In the second row, there will be 8 such unique variables, as 8 entries are to the right of the diagonal.
In the ninth row, there will be 1 such unique variable, as only one entry is to the right of the diagonal.
In the last row, there will be no entry to the right of the diagonal.

In a nutshell, we have to find the number of entries which are to the right of the diagonal. This will be $9+8+7+6+5+4+3+2+1+0=45$.

---

11. Question

Let $T:\mathbb{R}^3 \to \mathbb{R}^2$ be defined by $T(x,y,z)=(2x+3y,6z)$. Choose the correct options about T.

- ☐ The matrix of T is $\begin{bmatrix} 2 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$.
- ☐ A basis for the range of T is $\{(2,2)\}$.
- ☐ The matrix of T is $\begin{bmatrix} 2 & 0 \\ 3 & 0 \\ 0 & 6 \end{bmatrix}$.
- ☐ A basis for the null space of T is $\{(1,-\frac{2}{3},0)\}$.
- ☐ A basis for the range of T is $\{(2,3),(0,6)\}$.
- ☐ A basis for the null space of T is $\{(1,-\frac{2}{3},0),(0,0,6)\}$.

11.1. Answers :

- The matrix of T is $\begin{bmatrix} 2 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$.
- A basis for the null space of T is $\left\{\left(1,\frac{-2}{3},0\right)\right\}$.
- A basis for the range of T is $\{(2,3),(0,6)\}$.

11.2. Solution

Let us find the matrix representation of T with respect to the standard ordered basis.

$$\begin{aligned} T(1,0,0) &= (2,0) \\ T(1,0,0) &= 2\cdot(1,0)+0\cdot(0,1) \end{aligned}$$

$$\begin{aligned} T(0,1,0) &= (3,0) \\ T(0,1,0) &= 3\cdot(1,0)+0\cdot(0,1) \end{aligned}$$

$$\begin{aligned} T(0,0,1) &= (0,6) \\ T(0,0,1) &= 0\cdot(1,0)+6\cdot(0,1) \end{aligned}$$

The matrix representation for T is $M=\begin{bmatrix} 2 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$. Hence, **option 1** is correct and **option 3** is incorrect.

The reduced row echelon form of M is $R = \begin{bmatrix} 1 & 3/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

The set of solutions for $Rx = 0$ will be the null space of T. Here, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

Converting $Rx = 0$ into linear equations, we get $x_3 = 0$, $x_1 = -\dfrac{3x_2}{2}$. Every vector in kernel of T will be of the form $\begin{bmatrix} -(3x_2/2) \\ x_2 \\ 0 \end{bmatrix} = x_2 \begin{bmatrix} -3/2 \\ 1 \\ 0 \end{bmatrix}$. Notice that $\begin{bmatrix} 1 \\ -2/3 \\ 0 \end{bmatrix}$ is of the form $\dfrac{2}{3} \cdot \begin{bmatrix} -3/2 \\ 1 \\ 0 \end{bmatrix}$. Hence, **option 4** is correct and **option 6** is incorrect.

We can see that the 2nd column of R does not have a leading entry. The first has a leading entry and so does the third column. Therefore, we take the corresponding columns from M as a basis for the range of T. $\{(2,0),(0,6)\}$ can be a basis for T. Notice that $span\{(2,0),(0,6)\} = \mathbb{R}^2$. Any two linearly independent vectors from $\mathbb{R}^2$ can be a basis for the range of T.
The set $\{(2,3),(0,6)\}$ is a linearly independent set. Therefore, it can be a basis set for the range of T.
However, $\{(2,2)\}$ cannot span the entire range of T as there is only one vector in $\{(2,2)\}$.
Hence, **option 2** is incorrect and **option 5** is correct.